The Conductor of the SIC-POVM Moduli Field

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Abstract

Suppose you hold the squared magnitudes of a SIC-POVM fiducial vector in dimension d. They are real, they sum to one, and their squares sum to $2/(d+1)$. You want to know which number field they live in, exactly rather than approximately. That field is an abelian extension of the real quadratic field $F = \mathbb{Q}(\sqrt{(d+1)(d-3)})$, and this paper determines its conductor.

Three results carry the answer, and the third is new to this version.

At $d = 12$ the moduli generate a field of degree sixteen over $\mathbb{Q}$, signature $(8, 4)$, abelian of degree eight over F, with conductor $\mathfrak{p}_2^3\mathfrak{p}_3$ of norm 192 and exactly one infinite place ramified. The field is not totally real. An earlier draft said the opposite, having been misled by an integer relation that was fitting noise rather than recovering structure, and the test that separates the two is whether a relation's coefficient height holds steady as working precision rises. It is applied throughout.

Where the class number of F exceeds one, the class group is not carried by the moduli. Dimension sixteen is the smallest such dimension and it settles the question: the σ-coinvariant count at the Appleby modulus is 16 against a required $d/2 = 8$, and dividing by the class group restores it. The moduli field is the ray class field modulo the class group. At $d = 2048$ this gives degree 2^{21} over F, not 2^{27}.

Third: at $d = 2048$ the rule collapses to $\mathfrak{p}_2^{12}\infty_1$, and since 2 is inert there, that modulus is the principal ideal generated by $4096 = 16^3$. The conductor is a cube of sixteen. Section 7 records why that number is worth naming rather than passing over, what supports reading it as the trilattice's own shape, and what would refute the reading.

Keywords: SIC-POVM · moduli field · conductor · ray class field · Stark units · archimedean ramification · Belnap trilattice · Zauner conjecture · PARI/GP

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Contents

1 Introduction

Here is the problem in one sentence. A symmetric informationally complete positive operator-valued measure in dimension d is a set of d^2 unit vectors in $\mathbb{C}^d$ any two of which have the same absolute inner product, $|\langle\psi_j, \psi_k\rangle|^2 = 1/(d+1)$ for $j \neq k$. Put less formally: you fill complex projective space with d^2 points all equally far apart, and you ask whether it can be done. Zauner conjectured in 1999 that it can, in every dimension, and that the vectors can be taken as the orbit of a single fiducial ψ under the d^2 Weyl–Heisenberg displacement operators [1]. The conjecture is open.

Here is why it is a number theory problem. Appleby, Bengtsson, Flammia, McConnell, and Yard [2, 3, 4] established that the fiducial coordinates are algebraic numbers living in ray class fields of the real quadratic field $F = \mathbb{Q}(\sqrt{m_d})$, with m_d the squarefree part of $(d+1)(d-3)$. That ties SIC existence to the real-quadratic case of Hilbert's twelfth problem [7]. Find the fiducial and you have a generator; find the conductor and you know which field to look in.

That is the gap this paper fills. Knowing a fiducial lives in *some* ray class field is not knowing *which*. For any explicit construction the conductor is the thing you need, because it determines the ray class group, the number of characters, and whether the Stark machinery applies at all. This paper determines the conductor for the moduli, the real quantities $N_k = |\psi_k|^2$. The determination is empirical in the honest sense: calibrated against dimensions where the fiducial is known exactly, then applied to one where it is not.

The rule is stated in full in Section 5. Writing $\mathfrak{p}_2$ for the prime of F above 2, the moduli generate the ray class field of F modulo the class group, at the modulus

$$\mathfrak{f}_d \;=\; \mathfrak{p}_2^{v_2(d)+1} \cdot \prod_{p \mid d,\, p \text{ odd}} \; \prod_{\mathfrak{p} \mid p} \mathfrak{p} \cdot \infty_1,$$

with exactly one infinite place ramified.

Three features of this answer are more important than the formula.

The first is archimedean. Exactly one infinite place ramifies, so the moduli field is real at the fiducial embedding but not totally real. Stark's machinery in its standard form wants a totally real class field and refuses anything else; PARI's `bnrstark` reports nonzero r_2 and exits. The construction needs the mixed-signature form, as [8] maintained.

The second is the class group, invisible in every dimension where the fiducial is known exactly because all of those have class number one, and settled at $d = 16$ in Section 8. It costs six powers of two at $d = 2048$.

The third is what the rule produces at $d = 2048$, and it is the reason this version exists. There $d = 2^{11}$ has no odd prime divisor, so the formula collapses to $\mathfrak{p}_2^{12}\infty_1$; and 2 is inert in F, so $\mathfrak{p}_2 = (2)$ and the modulus is the principal ideal

$$\mathfrak{f}_{2048} \;=\; (2^{12})\,\infty_1 \;=\; (4096)\,\infty_1 \;=\; (16^3)\,\infty_1.$$

The conductor is generated by sixteen cubed. Section 7 treats that as a fact worth naming rather than an arithmetic accident, and is explicit about which part of the treatment is arithmetic and which is identification.

2 The moduli field is not the coordinate field

It is tempting to treat the moduli field and the coordinate field as interchangeable. They are not. For a fiducial ψ the coordinates ψ_k are complex; the moduli $N_k = |\psi_k|^2$ are real. But reality at the fiducial embedding does not imply total reality: the field can be, and is, complex at the conjugate place.

The practical teeth are that PARI's `bnrstark` computes Stark units for totally real class fields and refuses anything else. It refuses the moduli field. An earlier version of this paper claimed the opposite on the strength of a computation that did not survive scrutiny. Here is what survived.

Proposition 2.1 ($d = 12$ moduli field)**.** *The twelve moduli of the $d = 12$ fiducial generate a field of degree sixteen over $\mathbb{Q}$ of signature $(8, 4)$. Individually they have degree 4 or 8; the minimal polynomial of N_0 is $97344\,x^4 - 32448\,x^3 + 3224\,x^2 - 104\,x + 1$. The field is abelian of degree eight over $F = \mathbb{Q}(\sqrt{13})$, with conductor $\mathfrak{p}_2^3\mathfrak{p}_3$ of norm* 192, one *infinite place ramified, ray class group $(\mathbb{Z}/2)^3$ of order eight, and trivial subgroup.*

The field is therefore not totally real, and the moduli field is the full ray class field at a modulus whose archimedean part is one place: not none, and not both.

Establishing this takes more care than it appears, and the care is the real content. An integer relation computed against a power basis will happily report membership in a field that does not contain the element, if the working precision gives the lattice reduction enough slack to fit noise. The discriminating test is not the residual, since small residuals are easy. The test is the height. A genuine algebraic relation keeps its coefficient height fixed as working precision rises; a fitted relation sees its height grow with the precision, because each new digit demands a new coefficient to match it.

At $d = 12$, N_1 has a degree-eight minimal polynomial of height 33 bits, unchanged at 600, 900, and 1300 digits. The primitive element $\sum_k kN_k$ has a degree-sixteen relation of height 80 bits, likewise unchanged. But the claim that N_1 lies in a totally real octic, which is the claim the earlier draft made, is returned at every one of the eight embeddings with heights of 177, 287, and 435 bits at those three precisions. That is the signature of a fit, not a fact. The moduli have degree at most eight each, they do not lie in a common octic, and together they generate the degree-sixteen field above.

A related trap governs the input. The exact coordinates are known to roughly sixteen hundred digits, so seeking a relation at twenty-five hundred is fitting digits that are not there. Every claim in this paper is stated at a precision the input supports.

3 How each field was identified

The method is the same at each calibration dimension and is stated once.

Take the fiducial to the precision the published data supports. Form the moduli $N_k = |\psi_k|^2$. Seek a minimal polynomial for each by integer relation against a power basis, and seek one for a primitive element, taken here as $\sum_k kN_k$. Accept a relation only if its coefficient height is stable across at least three working precisions spanning several hundred digits. Identify the field from the accepted polynomial, compute its signature, and verify abelian-ness over

F by checking that the degree divides the ray class group order at a candidate modulus and that the Galois group is what the ray class group says it should be.

The conductor is then read off rather than guessed: enumerate moduli of F of small norm, compute the ray class group at each with the infinite places treated both ways, and select the one whose group has the right order and type and whose subgroup is trivial. Where several moduli agree the smallest is taken, which is what conductor means.

Two warnings, both learned rather than anticipated. The height test above is necessary and not decorative. And the archimedean part must be searched, not assumed: taking both infinite places unramified because the moduli are real is exactly the error that produced the retracted claim.

4 The three calibration points

Three dimensions have fiducials known exactly and class number one: $d = 4, d = 8, d = 12$. All three were computed by the method of Section 3.

Dimension four has $v_2(4) = 2$ and receives conductor exponent three. Dimension eight has $v_2(8) = 3$ and receives exponent four. Dimension twelve has $v_2(12) = 2$ and receives exponent three, together with the prime above three, since three divides twelve.

The reason dimension eight was worth the effort is that it refutes the naive guess. A constant exponent of three fits dimensions four and twelve perfectly well. Only $d = 8$ separates the constant from $v_2(d) + 1$, and it does so decisively.

The odd part carries a subtlety worth appreciating. Three splits in $\mathbb{Q}(\sqrt{13})$, so the factor appearing in the $d = 12$ conductor has norm three, while three is inert in $\mathbb{Q}(\sqrt{5})$ with norm nine. The odd contribution is a statement about primes of F lying above primes dividing d, not about rational integers. The two cases would be indistinguishable if only one had ever been computed, which is another way of saying that three calibration points is the minimum rather than a luxury.

5 The rule

Conjecture 5.1 (Conductor of the moduli field)**.** *Let d be even, $F = \mathbb{Q}(\sqrt{m_d})$ with m_d the squarefree part of $(d+1)(d-3)$, and $\mathfrak{p}_2$ the prime of F above 2. The moduli $N_k = |\psi_k|^2$ of a Zauner-symmetric Weyl–Heisenberg fiducial in dimension d generate the ray class field of F at the modulus*

$$\mathfrak{f}_d = \mathfrak{p}_2^{v_2(d)+1} \cdot \prod_{p|d,\ p \text{ odd}} \prod_{\mathfrak{p}|p} \mathfrak{p},$$

with exactly one infinite place ramified, taken modulo the class group of F. Its degree over F is $|\mathrm{Cl}_{\mathfrak{f}_d}|/h(F)$. The field is real at the fiducial embedding and not totally real.

The formula is compact and every piece earns its place. The 2-adic exponent is forced by $d = 8$ against $d = 4$ and $d = 12$. The odd part is forced by the splitting behaviour just described. The archimedean part is forced by Proposition 2.1. The quotient by the class group is forced by $d = 16$, which is the subject of Section 8.

6 Dimension 2048

Two thousand and forty-eight is the eleventh power of two. It has no odd prime divisors, so Conjecture 5.1 collapses to $\mathfrak{f}_{2048} = \mathfrak{p}_2^{12}\infty_1$. Here $m_d = 4190205 = 3 \cdot 5 \cdot 409 \cdot 683$, the class number is 64 with class group $[32, 2]$, and two is inert in F since $m_d \equiv 5 \pmod 8$.

Because two is inert, $\mathfrak{p}_2 = (2)$, and the modulus is principal:

$$\mathfrak{f}_{2048} \;=\; (2^{12})\,\infty_1 \;=\; (4096)\,\infty_1. \tag{1}$$

The ray class group at $\mathfrak{p}_2^{12}\infty_1$ has order 2^{27}, with cyclic type $[4096, 1024, 8, 4]$. The class number is 64, so by Section 8 the moduli field has degree 2^{21} over F and 2^{22} over $\mathbb{Q}$.

To see that this prediction is not sitting on a plateau where an error of one would be invisible, the neighbouring exponents were computed with the infinite places unramified, so that growth in the finite part stands on its own.

exponent k	$\lvert \mathrm{Cl}_{\mathfrak{p}_2^k} \rvert$	$\log_2$	cyclic type
9	2097152	21	$[1024, 128, 8, 2]$
10	8388608	23	$[2048, 256, 8, 2]$
11	33554432	25	$[4096, 512, 8, 2]$
12	67108864	26	$[4096, 1024, 8, 2]$
13	134217728	27	$[4096, 2048, 8, 2]$

The growth is a factor of four per step up to the eleventh power and a factor of two from there. The predicted exponent falls exactly at the change of régime. An error of one either way is an error of a factor of two or four in the degree, which is not subtle.

Read in valuations the filtration has a single generating shape, and Figure 1 shows it. Writing each cyclic invariant as its 2-adic valuation, every level from $k = 5$ upward is $[a,\ b,\ 3,\ 1]$: the tail is frozen and never moves again. The gap between the two leading invariants is $a - b = 3$, equal to the valuation of the third invariant, and it holds for the whole fourfold régime. Growth of four is therefore not two independent doublings but one invariant advancing while its distance to the next is pinned at the tail's valuation.

The geometry of that statement is a staircase with a ceiling. The leading invariant climbs one valuation per level, which is the ray class group acquiring one factor of two per step in the finite part of the modulus; the second invariant climbs in parallel three valuations behind, so the pair advances as a rigid body and the group grows by four. Nothing in the tail participates: from $k = 5$ the third and fourth invariants sit at 3 and 1 and never move, so the whole of the growth lives in a two-dimensional slice of a four-dimensional filtration. The régime changes exactly when the leader saturates at $12 = v_2(2d)$, which is a ceiling and not a coincidence: the leading invariant cannot exceed the 2-adic content of $2d$, so once it has absorbed all of it, only the second can still move. The rigid body stops being rigid. The gap collapses through $2, 1, 0$ at $k = 12, 13, 14$, the second invariant closes the distance and catches the first at $k = 14$, and at $k = 15$ the leader resumes. Growth is two throughout, because exactly one invariant is moving at a time.

The predicted exponent is the last level at which the gap is still pinned, the last moment before the structure changes character. It is therefore not a point chosen inside a smooth régime, where an error of one would be invisible; it is the corner itself.

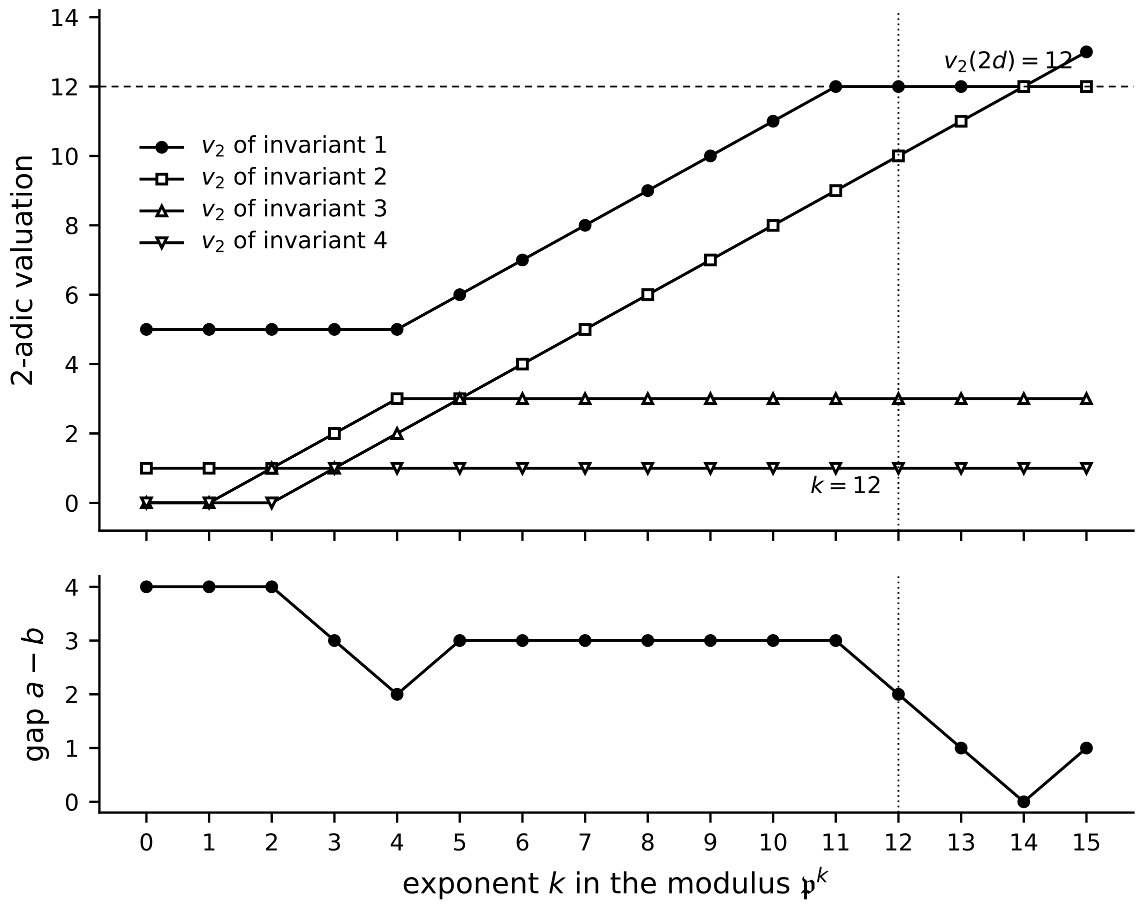

Figure 1: The 2-adic filtration of the moduli field at $d = 2048$, computed with PARI/GP and closed at $k = 15$. Above: the valuation of each cyclic invariant of $\mathrm{Cl}_{\mathfrak{p}_2^k}$. The tail is frozen from $k = 5$; the leading pair advances as a rigid body until the leader meets the ceiling $v_2(2d) = 12$. Below: the gap $a - b$ between the two leading valuations, pinned at the tail's valuation throughout the fourfold régime and collapsing through $2, 1, 0$ once the ceiling is reached. The predicted exponent $k = 12$ is the corner.

The calibration and the prediction are recorded in Lean 4 alongside the tower data, with the exponent rule and the fact that dimensions four and eight share a base field but differ in exponent stated as decidable propositions and discharged by evaluation [7]. The tower has since been carried to $k = 15$, which closes it: every ray class degree in the Lean 4 development is computed rather than extrapolated, and the halving is sustained, holding at exactly two for each of $k = 12, 13, 14, 15$.

7 The modulus is sixteen cubed

Equation (1) says the conductor at $d = 2048$ is generated by 4096. That number is 2^{12}, and it is also 16^3. This section says why the second reading is worth recording.

The programme this paper belongs to carries a logical substrate that is neither Boolean nor merely four-valued. Its base is the four-element set

$$I = \{T, F, t, f\}$$

of constructively proven, constructively refuted, non-constructively acceptable, and non-constructively rejectable. The trilattice is the full powerset $\mathcal{P}(I)$, so it has $2^4 = 16$ states, and it carries three genuinely distinct orderings:

$$x \leq_i y \iff x \subseteq y, \tag{2}$$
$$x \leq_t y \iff x \cap \{T, t\} \subseteq y \cap \{T, t\} \text{ and } y \cap \{F, f\} \subseteq x \cap \{F, f\}, \tag{3}$$
$$x \leq_c y \iff x \cap \{T, F\} \subseteq y \cap \{T, F\} \text{ and } y \cap \{t, f\} \subseteq x \cap \{t, f\}. \tag{4}$$

Information, truth, constructivity: sixteen states, three orders [5, 6]. The pair $(16, 3)$ travels with the structure.

The passage from the four to the sixteen deserves a statement, because it happens twice over and for a reason that does not generalise.

Proposition 7.1. $|\mathcal{P}(I)| = 2^4 = 16$ *and* $|I|^2 = 4^2 = 16$. *The equation* $2^n = n^2$ *has exactly two solutions in the positive integers,* $n = 2$ *and* $n = 4$.

Proof. Check $n = 1, 2, 3, 4$ directly: $2 \neq 1$, $4 = 4$, $8 \neq 9$, $16 = 16$. For $n \geq 5$ argue by induction that $2^n > n^2$. The base case is $2^5 = 32 > 25$. If $2^n > n^2$ with $n \geq 5$ then $2^{n+1} = 2 \cdot 2^n > 2n^2 \geq (n + 1)^2$, the last step being equivalent to $n^2 - 2n - 1 \geq 0$, which holds for $n \geq 3$. □

So four is the only place above the trivial case where enumerating all subsets and letting the base refer to itself give the same count. That is what licenses reading sixteen as *four self-applied* rather than merely *four enumerated*, and nothing more general licenses it.

Conjecture 7.2. *The* 16 *and the* 3 *in* $\mathfrak{f}_{2048} = (16^3)\infty_1$ *are the trilattice's sixteen states and three orderings: the conductor of the* $d = 2048$ *moduli field is the trilattice raised to its own number of orders.*

Three independent things support this, and none of them is inflated by saying so.

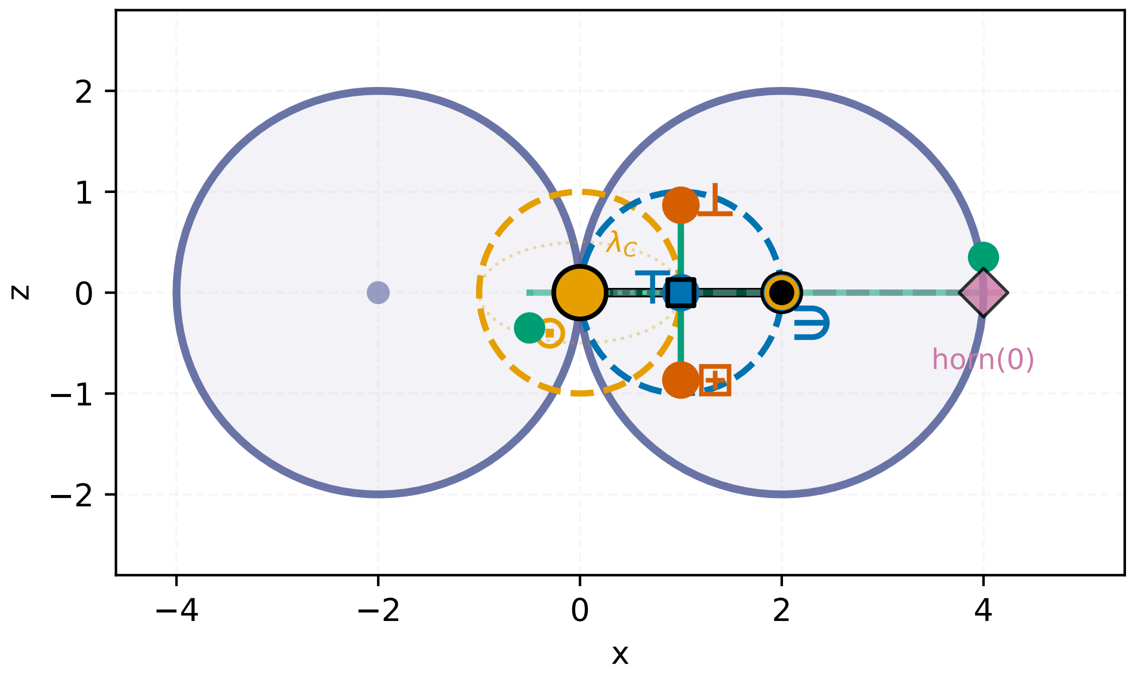

Figure 2: The side-on section, in the plane of the syzygy axis, drawn by the same generator that produces Figure 3. The ∈ shell is centred on ⊙; the evaluator sphere carries the same radius and is centred one radius along the axis, so ⊙ lies exactly on it and the two are tangent. The segment between the centres is the coupler. The evaluator trine is seen edge-on in this plane and stands at 120° only in the perpendicular one, which is why the two sections are drawn separately rather than combined.

The arithmetic is exact. Equation (1) is not an approximation and contains no tuned parameter. Either $\mathfrak{p}_2^{12}$ is principal and generated by 4096 or it is not, and inertness of 2 settles that it is.

The same pair is already present geometrically. Independently of any ray class computation, the evaluator arms of this programme trace a $(16,3)$ torus knot on the horn torus, sixteen toroidal windings against three poloidal ones, with bridge number three matching the three arms that carry the four-valued kernel. That pair was in the programme before this conductor was computed, and Figure 2 shows that pair of spheres in section.

Figure 3 shows the equator that claim is made on. The horn torus is the degenerate case $R = r$, where the tube radius equals the centre radius and the hole closes to a single point. That point is the pinch, and it is not a defect of the drawing: it is where the boundary meets itself, which is why the surface can carry a self-referential reading at all. The three evaluator arms sit at 120° on the equator, the A_2 geometry, and the three poloidal windings of the $(16,3)$ knot are those same three arms seen as a circuit rather than as a triple. The sixteen toroidal windings are the trilattice's sixteen states carried once around. Nothing here is scaled for legibility: $R = r$ exactly, so the two lobes of the side-on view touch rather than overlap, which is the whole content of the pinch.

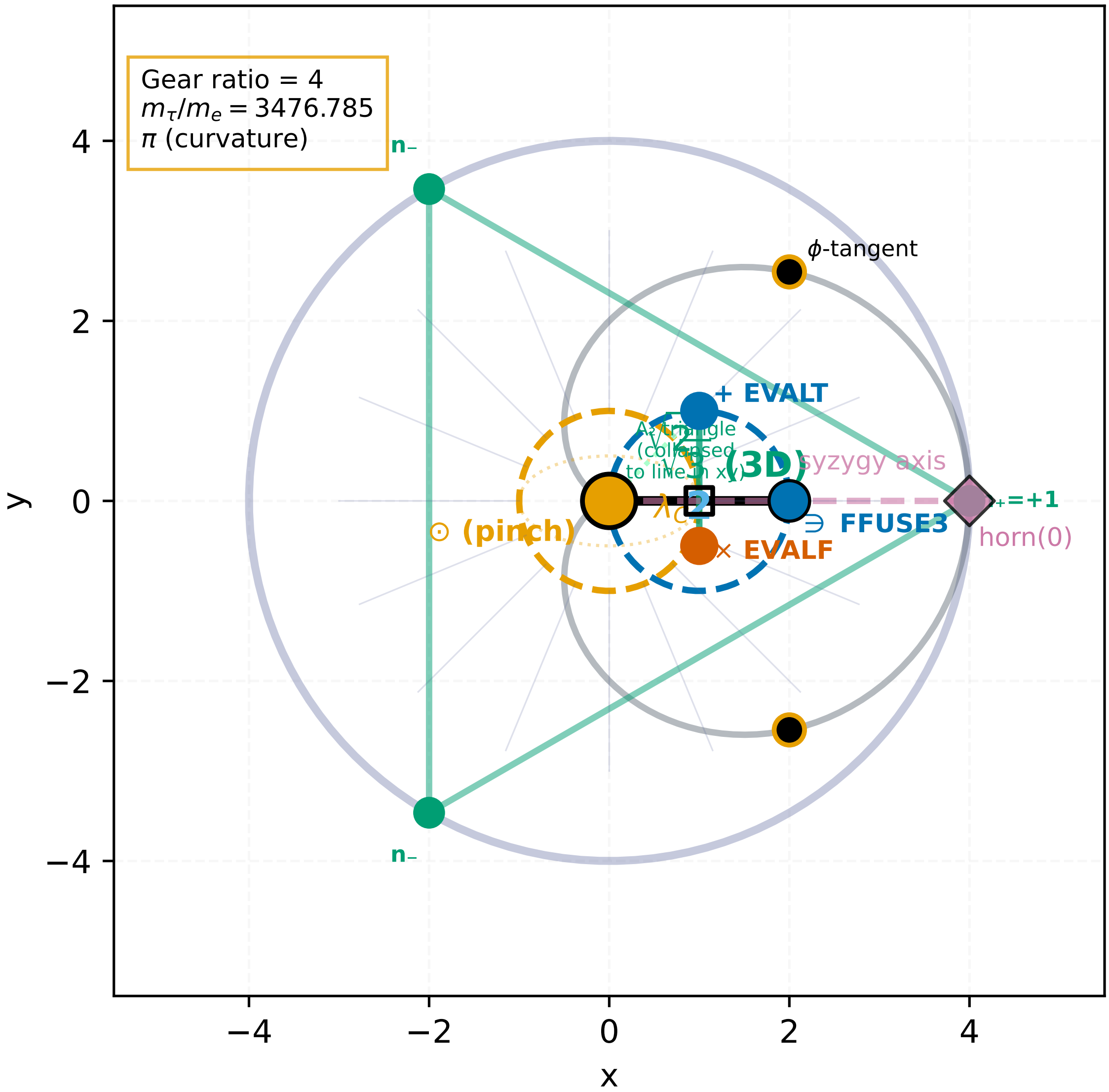

Figure 3: The equatorial section of the horn torus, drawn at $R = r$. The pinch ⊙ sits at the origin, where the hole has closed to a point and the boundary meets itself. The three evaluator arms stand at 120° and span the A_2 triangle; they are the three poloidal windings of the $(16, 3)$ knot, seen as a circuit rather than as a triple. Rendered from the generator that draws the kernel geometry, in its print theme.

The exponent and the calibrating dimension coincide. The leading cyclic invariant saturates at $12 = v_2(2d)$, which is also the conductor exponent and also the dimension at which the four-valued base is carried on twelve axes. We note the coincidence and deliberately do not build on it; three occurrences of the integer twelve in one programme is the kind of thing that wants a reason before it wants a use.

Remark 7.3 (What would refute it)**.** *The reading fails if the conductor at other $d = 2^n$ is not a power of sixteen. The rule gives $\mathfrak{f}_{2^n} = \mathfrak{p}_2^{n+1}\infty_1$, principal whenever 2 is inert, so the generator is 2^{n+1} and is a power of sixteen exactly when $4 \mid n + 1$. That happens at $n = 3, 7, 11, 15$, giving $d = 8, 128, 2048, 32768$ with generators $16, 16^2, 16^3, 16^4$. So the claim is not that every dimension has a sixteen-power conductor; it is that the sixteen-power dimensions form* *the arithmetic progression $n \equiv 3 \pmod 4$, and that $d = 2048$ is the third of them. If the $d = 8$ conductor were not generated by 16, or if the $(16, 3)$ knot invariance failed somewhere in that progression, or if the three orderings of Equations (2) to (4) proved dependent so that the structure is a bilattice, the reading should be dropped.*

Remark 7.4. *Dimension eight is the first member of that progression, and it is one of the three calibration points where the fiducial is known exactly. Its conductor is $\mathfrak{p}_2^4$, and $m_d = 5$ there with 2 inert, so the generator is 16 itself. The progression therefore begins at a dimension where the answer is not predicted but computed, which is the strongest position the claim can be in without the family-wide computation.*

8 The class group, settled at dimension sixteen

Every ray class field contains the Hilbert class field. So a rule naming the full ray class field and a rule naming its quotient by the class group differ by a factor of $h(F)$. All three calibration dimensions have $h = 1$, where the two agree, so nothing in Section 4 tells them apart. To see the class group you need a dimension where it is nontrivial, and dimension sixteen is the smallest.

At $d = 16$ we have $m_d = 221 = 13 \cdot 17$, class number two, two inert in $F = \mathbb{Q}(\sqrt{221})$, and $v_2(16) + 1 = 5$. The modulus is $\mathfrak{p}_2^5\infty_1$, the ray class group has order 256 with cyclic type $[16, 8, 2]$, giving 128 over F after the class group is divided out. The 2-power tower below is listed with the infinite places unramified, so that growth in the finite part is visible on its own.

k	$\lvert \mathrm{Cl}_{\mathfrak{p}_2^k} \rvert$	cyclic type	narrow order
3	16	$[8, 2]$	64
4	64	$[16, 4]$	256
5	128	$[16, 8]$	512
6	256	$[16, 16]$	1024

But the deciding quantity is not the order of this group. It is the count of independent moduli the group has to carry. Galois conjugation by the nontrivial automorphism σ of F pairs the moduli, $N_{k+d/2} = \sigma(N_k)$, so a moduli field must support exactly $d/2$ of them, and that count is read off the σ-coinvariant quotient of the ray class group at the Appleby modulus $(3d)$.

Proposition 8.1 (The coinvariant count at $d = 16$)**.** *Write G_d for the ray class group of F at $(3d)$ with both infinite places unramified, and G_d^σ for its σ-coinvariants. Then $\lvert G_{16}^\sigma \rvert = 16$, which is not $d/2 = 8$, while*

$$\frac{\lvert G_{16}^\sigma \rvert}{\lvert \mathrm{Cl}(F)^\sigma \rvert} = \frac{16}{2} = 8 = \frac{d}{2}.$$

The same identity holds at $d = 4$, 8, 12, where the class number is one and the denominator is trivial, so those dimensions say nothing either way. It is $d = 16$ that carries the content.

This is not a law for every dimension, and the reason is arithmetic rather than accidental, so it is worth setting down. It holds at $d = 4, 8, 12, 16, 24, 32, 36$ and fails at $d = 20, 28, 40$. Quoting the corrected count throughout, the coinvariant order divided by $\lvert \mathrm{Cl}(F)^\sigma \rvert$, the three failures give 8, 12, and 16 against $d/2$ of 10, 14, and 20.

The dividing line is the odd part of $d/2$. For an odd prime q, a ray class group at a modulus divisible by $\mathfrak{q}$ but not $\mathfrak{q}^2$ has no q-torsion: the local unit group has order $N\mathfrak{q} - 1$, equal to $q - 1$ when q splits and $q^2 - 1$ when q is inert, and q divides neither. The q-torsion first appears at $\mathfrak{q}^2$. The modulus $(3d)$ supplies 3^2 exactly when $3 \mid d$, and supplies no other odd prime squared unless that square already divides d. So the count reaches $d/2$ when the odd part of $d/2$ is a power of three, and falls short otherwise. Supplying the square by hand does not repair it: at $d = 20$ the moduli 25 and 100 give counts 20 and 40 where 10 is wanted; at $d = 28$ the modulus 49 gives 21 where 14 is wanted.

So Proposition 8.1 is a statement about dimension sixteen and not an instance of a general law, and it should be weighed as one dimension. What it says there is nonetheless exact: the raw count overshoots by precisely the class number, and dividing by the class group restores the number of independent moduli that Galois pairing requires.

Together with the typing of the two readings, where the reading that keeps the class group grounds itself on a non-abelian Galois group that the ray class group $\mathbb{Z}/16 \times \mathbb{Z}/8$ does not have, the conclusion is clean. The class group is not carried by the moduli. The moduli field is the ray class field at $\mathfrak{f}_d$ modulo the class group, of degree $|\mathrm{Cl}_{\mathfrak{f}_d}|/h(F)$ over F: degree 128 at $d = 16$, and $2^{27}/64 = 2^{21}$ at $d = 2048$. The calibration of Section 4 is untouched, and what changes is only the reading of the word *full*. It changes the $d = 2048$ degree by six powers of two.

One further fact about $d = 16$ deserves a place in the record. The $d = 16$ SIC is in hand numerically: the Zauner eigenspaces have dimensions five, six, and five; only the six-dimensional sector carries fiducials; and eight of them, separated by their moduli multisets, refine against the full overlap system past two thousand digits. The sum of moduli and the sum of their squares are exact to precision.

That computation also fixes the limit of recognition from below. With the moduli field of degree 256 over $\mathbb{Q}$ at $d = 16$, an individual modulus has no minimal polynomial within reach of integer relation. Indeed $\sqrt{221}$, which certainly lies in the field, is not recovered from the power basis of a modulus to degree twenty at twenty-five hundred digits. The height-collapse signature that identifies the canonical fiducial at $d = 8$, where the field has degree eight, does not survive the growth of the field. The route at $d = 16$ and beyond is from the field down, not from the numbers up.

9 The tower data, and what asserting it concealed

The tower entered LEAN 4 as thirty axioms: the degree at each of sixteen levels, the 2-adic valuations, the leading and trailing cyclic invariants, the class number, the narrow degree, and three field types. That is an honest way to record computed field data, and it is what the artifact statement means by the boundary between what is proved and what is assumed. It is also more than the data requires.

The sixteen degrees are not independent. The same module states the two laws they obey, a ratio of four through the maximal-growth interval $3 \le k \le 11$ and of two once the leading invariant saturates, so only the three seed values 64, 64, 128 are data. Written as a recursion on those, every level is decidable and the growth law becomes a consequence rather than a companion assumption. The valuations follow, though not by decision: ν_2 is `padicValNat`, which is noncomputable, and each level is instead an instance of $\nu_p(p^n) = n$, available because every degree in the tower is a power of two. The narrow degree is four times the wide one, the factor this manuscript already attributes to the two real places, which turns the assertion 2^{28} at $k = 12$ into a theorem. The class number is a value, returned independently by the kernel's own tower and Rédei distillations, the latter with 4-rank one matching $[32, 2]$.

The three types were declared `Type 0` with no inhabitants: placeholders for constructions Lean 4 does not have. Constructing class fields is not what is wanted here. Each is recorded instead as the twelve-slot type it occupies, the narrow field differing from the wide in the parity slot alone, since ramification at the real places is a parity condition. Nothing is claimed of them beyond position.

Thirty axioms become none, and the module still carries no `sorry`. The result of interest is not the count. Discharging the tail invariant showed it was false. It asserted that the last two cyclic invariants are $(8, 2)$ for $4 \leq k \leq 15$; the tower table printed in the same module gives $k = 4$ as $[32, 8, 4, 2]$, whose last two are $(4, 2)$. Stabilisation begins one level later. The bound was off by one against data sitting a few lines above it, and as an axiom it could have stayed there indefinitely, since an axiom is consistent with its own table by fiat. As a theorem it failed immediately and named the level it failed at.

That is the argument for spending the effort. Reducing assumption count is tidiness; what the reduction buys is that assertions which contradict the record announce themselves.

10 What follows

The prediction is checkable without solving the $d = 2048$ SIC. Computing the ray class field at the predicted modulus and confirming that its degree, signature, and Galois structure are consistent with carrying $d/2$ independent moduli under the pairing induced by the nontrivial automorphism of F is a computation in the field alone, with no fiducial required. The flat autocorrelation conditions on the moduli then serve as an independent check, and the condition that the sum of squares equals $2/(d + 1)$ has already been verified numerically in dimension eight to high precision from a fiducial obtained with no algebraic input at all.

Dimension twenty no longer needs reconciling. Its coinvariant count falls short because 5-torsion is absent from every modulus that keeps the two-part in range, as Section 8 sets out. What it costs is the generality of the count, not the $d = 16$ reading.

Conjecture 7.2 suggests one further computation, and it is cheap. The sixteen-power dimensions are $d = 2^n$ with $n \equiv 3 \pmod 4$, of which $d = 8$ is already computed and $d = 128$ is the first untested. Confirming that the $d = 128$ conductor is $\mathfrak{p}_2^8 \infty_1$ with 2 inert, so that its generator is 16^2, would put two computed points on the progression instead of one. That is a ray class computation in a single field and it does not require a fiducial. A failure there would settle the matter in the other direction, which is the useful property of the test.

Artifact statement

The field computations were performed in PARI/GP [9]. The $d = 16$ settlement is formalized in Lean 4 as its own module, sorry-free, carrying the tower data, the coinvariant count, the abelian type of the ray class group, and the propagation of the correction to $d = 2048$. It and the Lean 4 calibration module live in the `p4ramill` library of `p4rakernel` (`https://github.com/umpolungfish/p4rakernel`), frozen on branch `crystalline/sic-moduli-conductor-2026-07-25` under tag `crystalline-sic-moduli-conductor-v1`, with `main` continuing. The tower discharge of Section 9 is frozen alongside it on branch `crystalline/d2048-axiom-free-2026-08-01` under tag `crystalline-d2048-axiom-free-v1`, where the moduli module carries zero axioms and zero sorries. Running `./verify_sic_moduli.sh` at the root of that repository elaborates the modules through the kernel and prints, dimension by dimension across $d = 2, 4, 8, 12, 16, 20, 2048$, what was checked, followed by the axiom dependency of each headline theorem, so that the boundary between what is proved and what enters as computed field data is visible in the output rather than left to the reader to reconstruct. This manuscript lives in `ig-docs` (`https://github.com/umpolungfish/ig-docs`). Every field computation reported here is reproducible from the PARI/GP calls named in the text: the modulus is read off with `rnfconductor`, and the candidate algebraic relations with `algdep`.

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The author would like to thank Harry T. Larson for imparting the importance of catching rising problems and never letting them go [10]. The conductor was found by refusing to accept a tool's refusal at face value: when `bnrstark` declined the coordinate field for having nonzero r_2, the question became which field the moduli actually generate — and that question had an answer.

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